

1.5 SOLUTION SETS OF LINEAR SYSTEMS

Homogeneous Linear Systems

A system of linear equations is said to be homogeneous if it can be written in the form $Ax=0$, where A is an $m \times n$ matrix and 0 is the zero vector in $\mathbb{R}^m$. Such a system $Ax=0$ always has at least one solution, namely, $x=0$ (the zero vector in $\mathbb{R}^n$). This zero solution is usually called the trivial solution. For a given equation $Ax=0$, the important question is whether there exists a nontrivial solution, that is, a nonzero vector x that satisfies $Ax=0$.

- The Existence and Uniqueness Theorem in Section 1.2 (Theorem 2) leads immediately to the following fact:
The homogeneous equation $Ax=0$ has a nontrivial solution if and only if the equation has at least one free variable.

Example 1: Determine if the following homogeneous system has a nontrivial solution. Then describe the solution set.

$$\begin{aligned} 3x_1 + 5x_2 - 4x_3 &= 0 \\ -3x_1 + (-2x_2) + 4x_3 &= 0 \\ 6x_1 + x_2 - 8x_3 &= 0 \end{aligned}$$

Solution: $x_1 \begin{bmatrix} 3 \\ -3 \\ 6 \end{bmatrix} + x_2 \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} -4 \\ 4 \\ -8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$